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Machine learning dataset and benchmark for accurate T cell receptor–pHLA binding prediction

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A central challenge in immunology and therapeutic design is accurately predicting the diverse interactions between T cell receptors and peptide–human leukocyte antigen (pHLA) complexes. Existing machine learning tools are hindered by incomplete sequence data and biased nonbinding examples. Here we present Hi-TPH, a large-scale hierarchical dataset featuring an on-the-fly selection method for generating nonbinding data. We further develop Hi-TPH-PLMs, a collection of protein language models (PLMs) with varied architectures and scales, fine-tuned on Hi-TPH. These models achieve a 19.2% performance gain over state-of-the-art tools on an external wet-lab test set. Leveraging the hierarchical structure of Hi-TPH, detailed analyses dissect the contribution of different molecular components to binding prediction and reveal their synergistic interplay; for instance, the prediction contribution of HLA relies on the presence of full T cell receptor chains. Hi-TPH and Hi-TPH-PLMs are publicly released to support the development of more reliable tools for advanced immunoinformatics research and personalized immunotherapy.

The ability to accurately predict the binding between a T cell receptor (TCR) and a peptide–human leukocyte antigen (pHLA) complex is a cornerstone of modern immunology and a critical enabler for next-generation therapeutics[1–4]. However, experimentally characterizing the vast repertoire of these interactions is a formidable task, limited by the immense diversity of TCRs, peptides and HLAs, as well as high costs and long timelines. Consequently, the field has increasingly embraced computational approaches, particularly advanced machine learning models[5–8]. These in silico tools enable rapid and large-scale analysis, a capability that is crucial for accelerating cancer vaccine development[9–12] and predicting immunotherapy efficacy[13,14].

Despite rapid advances in TCR–pHLA binding prediction models, their performance and practical utility are fundamentally constrained by two data-centric challenges: incomplete sequence data and biased nonbinding TCR–pHLA pairs[15,16]. A vast majority of existing datasets focus predominantly on the complementarity-determining region 3 β (CDR3β) loop of TCR[17–19]. Omitting other components of TCR–pHLA hinders the development of comprehensive prediction models and the understanding of prediction contribution[20,21], as compelling biological evidence suggests that the CDR3α loop and full TCR variable chains can be decisive in determining binding specificity[20–22]. Common practices rely on fixed nonbinding data along the whole training process, which can introduce profound biases, leading to models that learn artifacts of the negative set rather than the true biophysical principles of nonbinding.

To address these challenges, we introduce Hi-TPH, a large-scale, meticulously curated and hierarchically structured dataset for TCR–pHLA binding prediction, focusing exclusively on HLA class I (referred to as HLA hereafter). Hi-TPH is stratified into four levels of progressively richer molecular components (Fig. 1), from basic peptide/CDR3β

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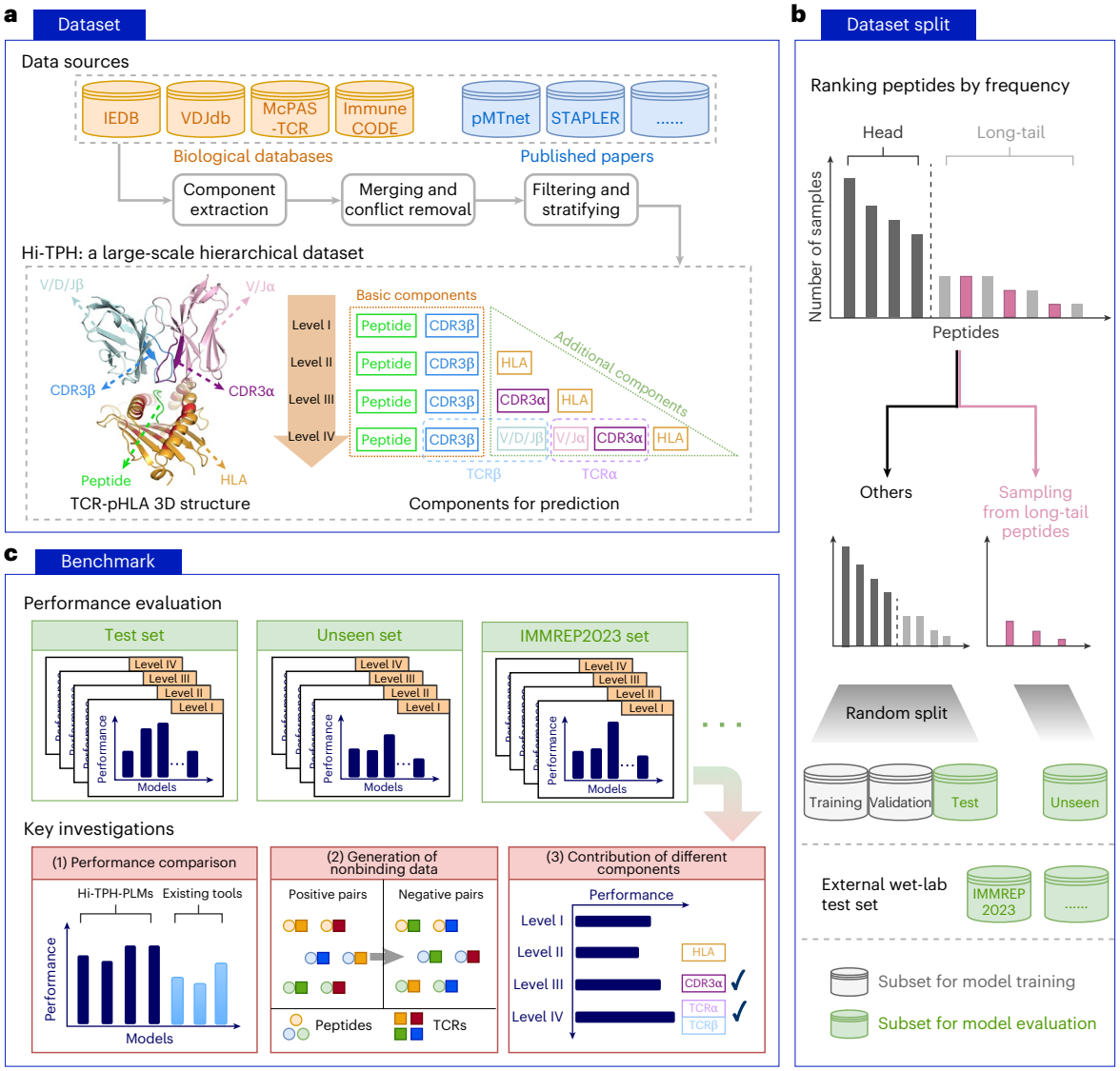


Fig. 1 | Overview of the Hi-TPH dataset construction, structure and benchmarking strategy for TCR–pHLA binding prediction. a, Hi-TPH dataset construction. Data from key biological databases (for example, IEDB and VDJdb) and published literature were curated via component extraction, conflict removal and stratification to create Hi-TPH. This large-scale dataset is organized into four hierarchical levels (I–IV), incorporating increasing TCR–pHLA components: level I (peptide, CDR3β), level II (adding HLA), level III (peptide, CDR3β, CDR3α and HLA) and level IV (peptide, HLA, full TCR α/β chains with V(D)J gene annotations). An illustrative three-dimensional (3D) structure depicts the TCR–pHLA interaction. **b**, Dataset splitting. Peptides exhibit a long-tail frequency distribution concerning their binding TCRs. Data are partitioned such that peptides from the long tail are randomly sampled to constitute the Unseen set, while the remaining peptides are randomly allocated into training, validation, and test sets. IMMREP2023 serves as an external test set with wet-lab validated binding labels. **c**, Key investigations on the benchmark: (1) the performance comparison of existing tools and Hi-TPH-PLMs (a collection of PLMs fine-tuned on Hi-TPH) on the external wet-lab validated IMMREP2023 set, (2) the effect of on-the-fly selection of nonbinding data and (3) the contribution of different prediction components to model performance.

sequences (level I) to comprehensive full TCR α/β chains with HLA data and V(D)J gene annotations (level IV). Each level of Hi-TPH is the most comprehensive and largest publicly available collection of its kind. At its most feature-rich level, Hi-TPH level IV represents a nearly sixfold expansion over previously available resources with full TCR chain information. Furthermore, Hi-TPH uses an on-the-fly selection strategy for nonbinding data generation to prevent the inherent biases of static ones during model training.

Owing to the expanded data scale, we further develop Hi-TPH-PLMs on the Hi-TPH dataset, a suite of protein language models (PLMs) with varied architectures and scales. These models were fairly benchmarked against existing tools on the wet-lab validated IMMREP2023 set[23], which can improve the binding prediction accuracy by 19.2% regarding the area under the receiver operating characteristic curve (AUROC). Some widely used tools falter on IMMREP2023 with performance near the level of random chance, due to limited data coverage and weak generalization to unseen peptides[18,23]. Besides, the performance degradation on unseen peptides is most pronounced when models rely solely on peptide and CDR3β information. These observations reinforce the critical need for datasets like Hi-TPH with broader peptide coverage and more complete molecular components. Benchmarks also confirm the superiority of on-the-fly nonbinding data generation over conventional fixed negative sets, which mitigates dataset biases and fosters the learning of biophysical principles.

Beyond the development of state-of-the-art (SOTA) models, the unique hierarchical architecture of Hi-TPH, along with the expanded scale at all levels, facilitates systematic analysis of how different molecular components contribute to binding prediction.

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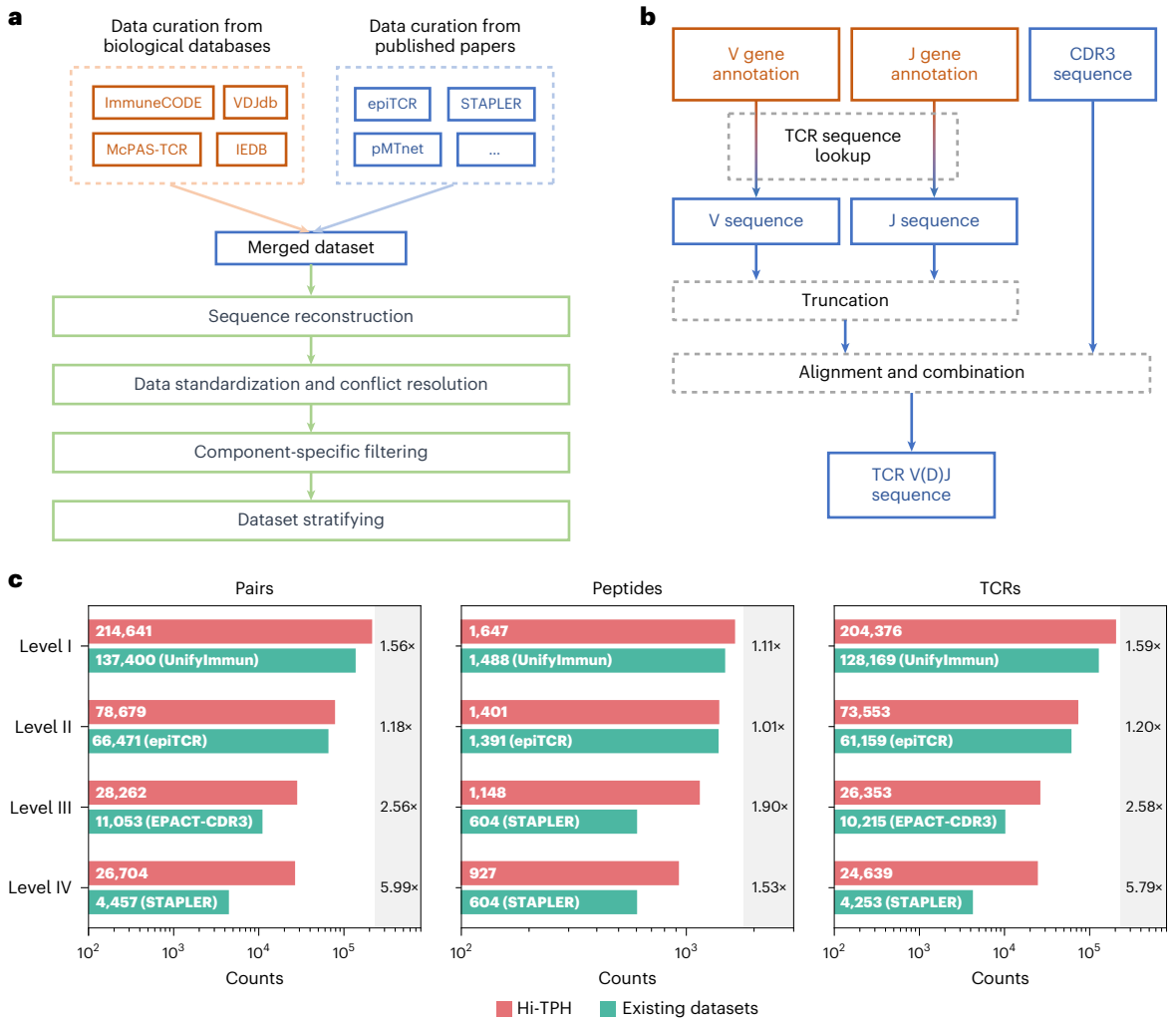


Fig. 2 | Data collection workflows and statistical characteristics of the Hi-TPH dataset. a, Schematic of the positive data collection pipeline. This workflow initiates with curating TCR–pHLA binding instances from biological databases (ImmuneCODE, VDJdb, McPAS-TCR and IEDB) and published studies (epiTCR, STAPLER, pMTnet and so on). Subsequent steps include sequence reconstruction, data standardization, conflict resolution, component-specific filtering and dataset stratifying. **b**, Detailed procedure for mapping V(D)J gene annotations to their corresponding full TCR V(D)J region sequences. This involves V and J gene annotation lookup, CDR3 sequence extraction and subsequent truncation, alignment and combination steps to reconstruct the complete V(D)J sequence. **c**, Comparative analysis of the Hi-TPH dataset size. The charts compare the number of TCR–peptide pairs, unique peptides and unique TCRs in Hi-TPH with the largest comparable public datasets across four levels. Counts are presented on a logarithmic scale.

Our investigation reveals a synergistic interplay among these components. For instance, incorporating the TCR CDR3α sequence provides a direct performance boost, while the contribution of the HLA sequence is context dependent. Its predictive value is negligible or even detrimental when paired only with TCR CDR3 loops, but becomes substantially positive once full TCR chains are included in the model. These findings provide crucial insights that can guide future model development and deepen our fundamental understanding of TCR–pHLA recognition.

This work offers the academic community a machine learning-oriented dataset, Hi-TPH, and a benchmarking platform to address persistent data-centric challenges in TCR–pHLA binding prediction. By fine-tuning PLMs on Hi-TPH, we establish a streamlined yet powerful approach that achieves superior generalization performance over existing SOTA tools. We demonstrate that this success is driven by key properties of Hi-TPH, including its on-the-fly nonbinding data generation and broader coverage of TCRs and peptides. In addition, the hierarchical structure of Hi-TPH provides a framework for dissecting the synergistic roles of individual molecular components. We anticipate that Hi-TPH will serve as a pivotal resource for the immunoinformatics community, accelerating the development of more reliable and clinically relevant prediction models.

Results

The Hi-TPH dataset

To confront the foundational data challenges, this study introduces Hi-TPH, a large-scale, actively maintained, and hierarchically structured dataset for TCR–pHLA binding prediction (Fig. 1a). The construction of Hi-TPH was a meticulous process involving the aggregation of binding data from biological databases (IEDB[24], VDJdb[25], McPAS-TCR[26] and ImmuneCODE[27]) and the curation of data from influential published studies[15,17]. This was followed by rigorous component extraction, conflict resolution and data stratification, resulting in a standardized and high-quality resource essential for the development of machine learning models (Fig. 2a). A key outcome of our processing pipeline was the standardized mapping of V(D)J gene annotations to full TCR V(D)J region sequences (Fig. 2b), yielding cleaner and more comprehensive input features for TCR–pHLA binding prediction.

The first major contribution of Hi-TPH is the sheer scale, which substantially surpasses existing publicly available datasets. As detailed

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Table 1 | Comparative overview of datasets for TCR–pHLA binding prediction

Dataset	Number of samples			Protein sequence						Gene		HLA type	Data collection	Year
	Pairs	Peptides	TCRs	Peptide	CDR3β	HLA	CDR3α	TCRα	TCRβ	V/Jα	V/D/Jβ			
DLpTCR[19]	7,121	304	6,583	✓	✓							HLA I	Curation	2021
PanPep[18]	32,080	699	29,467	✓	✓							HLA I	Curation	2023
TEIM[36]	45,481	355	44,227	✓	✓							HLA I	Curation	2023
DAISY[56]	70,491	127	64,166	✓	✓							HLA I	Curation	2025
UnifyImmun[28]	137,400	1,488	128,169	✓	✓							HLA I	Curation	2025
NetTCR-2.0[57]	11,431	17	11,425	✓	✓		✓*					HLA I	Curation	2021
pMTnet[13]	32,070	602	28,864	✓	✓	✓						HLA I	Curation	2021
epiTCR[17]	66,471	1,391	61,159	✓	✓	✓						HLA I	Curation	2023
PISTE[40]	32,508	607	29,687	✓	✓	✓						HLA I	Curation	2024
EPACT-CDR3[58]	11,053	240	10,215	✓	✓	✓	✓					HLA I	Curation	2024
EPACT[58]	9,936	179	9,265	✓	✓	✓	✓			✓	✓	HLA I	Curation	2024
TCRAI[59]	8,130	16	8,101	✓	✓	✓	✓			✓*	✓*	HLA I/II	Curation + Experiment	2021
ERGO-II[20]	27,260	389	26,548	✓	✓	✓	✓*			✓*	✓*	HLA I/II	Curation	2021
STAPLER[15]	4,457	604	4,253	✓	✓	✓	✓	✓	✓	✓	✓	HLA I	Curation	2023
Hi-TPH level I	214,641	1,647	204,376	✓	✓							HLA I	Curation	2025
Hi-TPH level II	78,679	1,401	73,553	✓	✓	✓						HLA I	Curation	
Hi-TPH level III	28,262	1,148	26,353	✓	✓	✓	✓					HLA I	Curation	
Hi-TPH level IV	26,704	927	24,639	✓	✓	✓	✓	✓	✓	✓	✓	HLA I	Curation	

This table summarizes characteristics of publicly available datasets alongside the Hi-TPH dataset, including sample counts (pairs, peptides and TCRs), presence of protein sequence features (Peptide, CDR3β, HLA, CDR3α, TCRα and TCRβ), gene annotations (V/Jα and V/D/Jβ), HLA type, data collection methods and year of publication for each resource. Statistics consider only positive (binding) TCR–pHLA pairs. '✓*' indicates partially available information. TCRα and TCRβ sequences denote full variable regions (V, D and J segments), excluding constant (C) regions.

in the comparative analysis below (Table 1 and Fig. 2c), Hi-TPH provides a high volume of data at every level of molecular detail. For instance, Hi-TPH level I, containing the basic peptide and CDR3β sequences, includes over 210,000 binding pairs, 50% larger than the next-largest comparable dataset[28]. At its most feature-rich level, the Hi-TPH level IV dataset represents a 5.99-fold expansion in paired TCR–pHLA interactions over any previously available dataset[15]. The increased scale of Hi-TPH enables the training of larger and deeper models, such as tuning PLMs for TCR–pHLA binding prediction.

Beyond its scale advantage, Hi-TPH also stands out for its comprehensiveness, encompassing a substantially broader spectrum of TCR sequences and peptide repertoires than existing datasets (Fig. 2c). This diversity is evident across its hierarchical levels. At level III, for example, Hi-TPH provides a 1.9-fold expansion in unique peptides over the next-largest comparable dataset[15]. Similarly, at its most feature-rich level IV, it offers a 5.8-fold expansion in unique TCRs[15], enabling models that leverage full-length TCR sequences to train on a much broader repertoire. This expanded coverage enables models to explore a more diverse molecular space during training, exposing them to a wider range of TCR–pHLA binding patterns.

Another key contribution of our dataset is the strategic and hierarchical structure. The dataset is deliberately stratified into four levels of progressively richer molecular information, a design that empowers systematic investigation into the relative contributions of diverse molecular components. This stratification ranges from level I, containing foundational peptide and TCR CDR3β sequences, to subsequent levels that incrementally add more data: level II incorporates the HLA sequence; level III combines peptide, CDR3β, CDR3α and HLA sequence; and level IV presents the most comprehensive feature set, including full TCR α and β chains, peptide, HLA and V(D)J gene annotations (Fig. 1a). This hierarchical design facilitates controlled in silico experiments, allowing us to systematically isolate the prediction contribution of each molecular component. This level of controlled analysis was previously intractable owing to the fragmented and inconsistent nature of available datasets.

Superior TCR–pHLA binding prediction with Hi-TPH-PLMs

To showcase the capacity of Hi-TPH to serve as a foundation for building advanced TCR–pHLA binding prediction models, we developed Hi-TPH-PLMs, a suite of PLMs fine-tuned on its large-scale, comprehensive and hierarchically structured data (Fig. 3). We then benchmarked Hi-TPH-PLMs against existing SOTA tools on the IMMREP2023 set[23]. As an external, wet-lab validated benchmark, IMMREP2023 provides a test of cross-dataset generalization. To avoid potential overlap between evaluation data and our model development data, we removed all identified overlapping TCR–pHLA pairs between IMMREP2023 and our Hi-TPH training/validation splits across all levels before reporting results.

Our findings clearly demonstrated the superiority of models trained on Hi-TPH. The best-performing Hi-TPH-PLM achieved a remarkable 19.2% relative improvement in AUROC over the top-performing existing tool on level III (Table 2 and Supplementary Fig. 1). By contrast, many established tools faltered on this difficult task, with several achieving AUROC scores near 0.5, the level of random chance. This result not only highlights a pervasive generalization challenge within the field but also showcases the performance boost conferred by training on our Hi-TPH data. Further analysis shows that identical PLMs outperform their counterparts trained on previous datasets (Supplementary Table 2) and retraining established tools on Hi-TPH also gains relative performance improvement (Supplementary Table 3). Moreover, Hi-TPH-PLMs remained among the top-performing models on the similarity-filtered IMMREP2023 subsets (Supplementary Tables 11 and 12) and outperformed retrainable baselines on the Hi-TPH test set (Supplementary Table 9). These results validate the core contribution of Hi-TPH to the performance advantage of Hi-TPH-PLMs.

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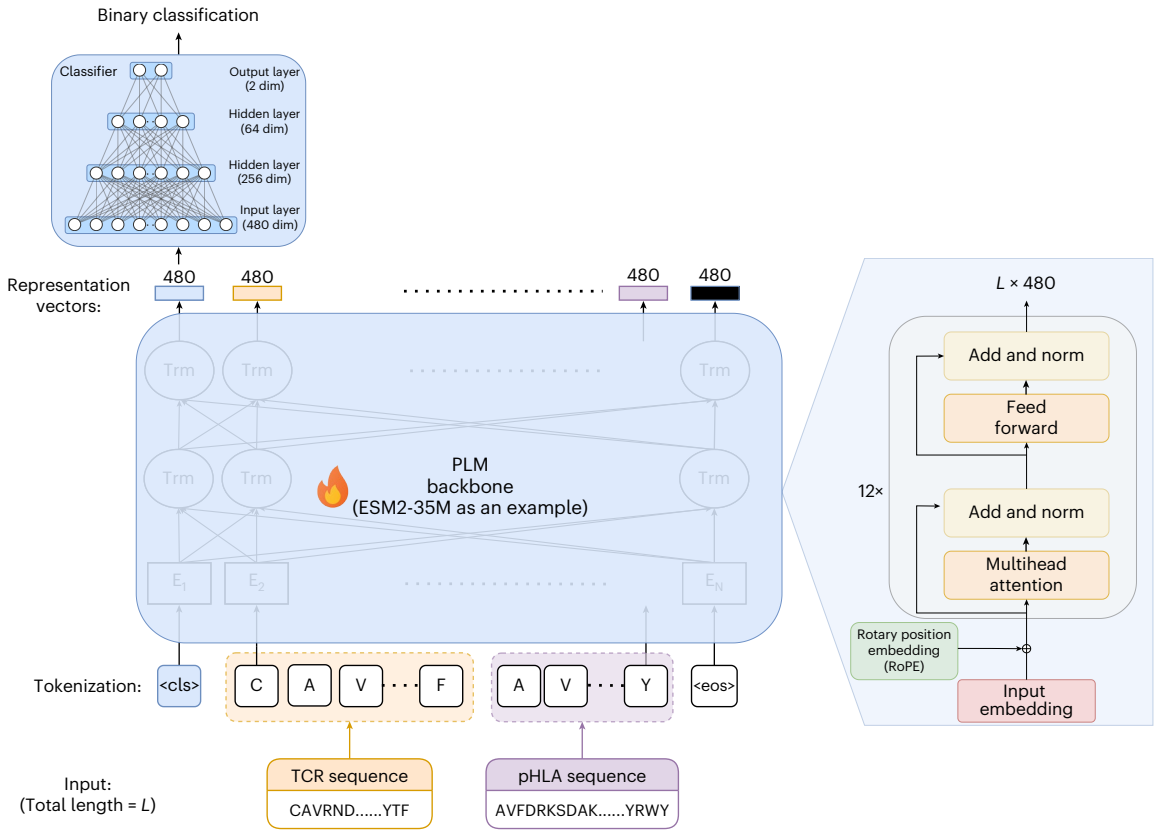


Fig. 3 | Architecture of Hi-TPH-PLMs. Given a TCR sequence and a pHLA sequence (peptide concatenated with HLA), we concatenate them into a single input sequence of total length L, prepend a classification token (<cls>), and append an end-of-sequence token (<eos>). The resulting sequence is fed into a pretrained PLM (ESM2 is shown as an example). We take the contextual embedding of the <cls> token (a d-dimensional (dim) representation, for instance, $d = 480$ for ESM2-35M) and feed it into a multilayer perceptron classifier to output the final binding prediction.

To pinpoint the cause of this generalization challenge, we conducted an internal analysis across our Hi-TPH-PLMs. Our results revealed a critical vulnerability: the performance degradation on unseen peptides was most severe for models relying solely on peptide and CDR3β sequences (Fig. 4a,b and Supplementary Table 1). This observation directly validates the core design of Hi-TPH. It confirms that the common practice of using basic prediction components is a key driver of generalization failure, thereby underscoring the necessity of Hi-TPH's approach: providing broader peptide coverage and incorporating the comprehensive components essential for building robust and clinically relevant prediction models.

Efficacy of on-the-fly selection for nonbinding data

A core feature of our Hi-TPH dataset is the integrated on-the-fly selection method for generating nonbinding data. We hypothesized that this dynamic approach, which creates new negative examples for each training batch, would outperform conventional methods that rely on a fixed set of nonbinding pairs. To validate this, we benchmarked the on-the-fly selection method against two prevalent fixed-set baselines (see 'On-the-fly selection method for nonbinding data generation' section in the Methods) across a wide range of conditions.

The superiority of the on-the-fly selection method is demonstrated in Fig. 4c,d, where it consistently outperforms both random and frequency-based fixed-set baselines regardless of the model scale, Hi-TPH data level or evaluation set (Supplementary Table 5). The benefit is particularly pronounced on the IMMREP2023 set, where on-the-fly selection of nonbinding data at level IV delivered relative AUROC improvements of 5.22% (ESM2-35M) and 4.80% (ESM2-150M) over the most competitive baseline.

This consistent advantage highlights the power of Hi-TPH's on-the-fly selection for nonbinding data generation. By exposing the model to a broader and more challenging set of nonbinders, this method effectively mitigates overfitting to static, potentially biased negative distributions[29], compelling the model to learn a more robust decision boundary. Thus, the on-the-fly nonbinding data generation is a core component of the Hi-TPH dataset that directly drives the development of superior prediction models.

Component contributions to TCR–pHLA binding prediction

Beyond enabling the training of SOTA models through its scale, comprehensiveness and on-the-fly nonbinding data generation, a core strength of Hi-TPH lies in its hierarchical design to enable a controlled analysis of component contributions. We trained Hi-TPH-PLMs on Hi-TPH level IV with different prediction components as inputs to dissect the specific contributions of TCR and HLA sequences to model performance.

There is a consistent performance improvement of Hi-TPH-PLMs on the IMMREP2023 set when progressively incorporating CDR3α and full TCR sequences (Fig. 4e and Supplementary Fig. 4a). Taking ESM2-35M as an example, adding the CDR3α sequence yielded an 18.2% relative gain regarding AUROC as compared with using peptide, HLA and CDR3β as model inputs. Extending the inputs from CDR3α and CDR3β to full TCR sequences further increased the improvement to 33.9%. These substantial gains indicate that comprehensive TCR sequence context captures critical determinants required for accurate binding prediction.

The contribution of the HLA sequence is context-dependent and varies with the completeness of the TCR sequence. Incorporating the HLA sequence causes a performance drop when taking peptide and

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Table 2 | Performance of TCR–pHLA binding prediction models, including existing tools and new models trained on our Hi-TPH dataset

Model	Level I		Level II		Level III		Level IV	
	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
PanPep[18]	0.5263	0.1635	–	–	–	–	–	–
TEIM[36]	0.5588	0.2586	–	–	–	–	–	–
UnifyImmun[28]	0.5072	0.1583	–	–	–	–	–	–
DAISY[56]	0.5194	0.1703	–	–	–	–	–	–
epiTCR[17]	0.5603	0.2435	0.5730	0.2442	–	–	–	–
$PISTE_{random}$[40]	–	–	0.5785	0.2025	–	–	–	–
$PISTE_{unipep}$[40]	–	–	0.5889	0.2850	–	–	–	–
ERGO-II[20]	–	–	–	–	0.5040	0.1718	0.5048	0.1720
STAPLER[15]	–	–	–	–	–	–	0.7391	0.5405
EPACT[58]	–	–	–	–	0.6011	0.3383	0.7500	0.5204
ESM2-8M	**0.6567**	0.3790	*0.5938*	0.3759	0.6770	0.3813	0.7404	0.5069
ESM2-35M	0.6290	0.3754	**0.5965**	0.3676	0.6740	0.4314	0.7574	*0.5493*
ESM2-150M	0.6372	0.3765	0.5867	0.3526	0.6828	0.4157	0.7399	0.4725
ESM2-650M	0.6235	0.3593	0.5748	0.3561	*0.7100*	0.4546	*0.7655*	0.5395
TAPE-BERT (38M)	0.6407	*0.3885*	0.5911	0.3740	0.6571	0.3935	0.7550	0.4839
AMPLIFY-120M-base	0.6138	0.3708	0.5878	0.3531	0.6711	0.4181	0.7165	0.5262
AMPLIFY-120M	0.6365	0.3774	0.5820	0.3707	0.6943	0.4278	0.7551	0.5259
AMPLIFY-350M	0.6309	0.3611	0.5790	0.3783	0.6996	*0.4671*	0.7617	0.5038
ProtBERT (420M)	0.6236	0.3657	0.5877	*0.3856*	0.6829	0.4244	0.7531	0.5141
Hi-TPH-PLMs-Avg.	0.6324	0.3726	0.5866	0.3682	0.6832	0.4238	0.7494	0.5136
Hi-TPH-PLMs-Ens.	*0.6508*	**0.4072**	0.5918	**0.3925**	**0.7163**	**0.4745**	**0.7979**	**0.5800**

Hi-TPH-PLMs-Avg. and Hi-TPH-PLMs-Ens. denote the mean performance across all Hi-TPH-PLMs and the performance of their ensemble, respectively (see 'Model Ensemble and Metric Averaging' section in the Methods). Performance was evaluated on the external IMMREP2023 set across four Hi-TPH levels. The comparison is measured by both AUROC and AUPRC. The highest score in each column is shown in bold, and the second highest is italicized.

CDR3β as model inputs (Fig. 4f, left, and Supplementary Fig. 4b). On the contrary, adding HLA consistently improves performance when full TCR sequences are available (Fig. 4f, right, and Supplementary Fig. 4b). These results indicate that the prediction model requires the information of CDR1 and CDR2 regions as context to interpret the effect of HLA in TCR–pHLA binding. This also aligns with established biophysical principles of TCR–pHLA recognition, where CDR1 and CDR2 loops interact with the HLA helices[30].

To further validate these findings, we conducted similar ablation studies on Hi-TPH levels II and III, where CDR3α and HLA exhibit opposite effects on the prediction performance (Supplementary Table 6). Noting that PLMs recognize interactions with attention weights, we further conducted attention map analyses (Supplementary Fig. 5), where the distribution of attention weights exhibits component-aware patterns reflecting similar component contributions and synergistic interactions.

Discussion

In this work, we introduced Hi-TPH, a large-scale dataset designed to address data-centric challenges in TCR–pHLA binding prediction. Hi-TPH offers a nearly sixfold expansion over previous resources with full TCR α/β chains and V(D)J gene annotations, largely mitigating the impact of incomplete sequence data. By employing an on-the-fly selection method, Hi-TPH mitigates the problem of nonbinding data bias. Hi-TPH-PLMs trained on Hi-TPH demonstrate consistent superiority on the external IMMREP2023 set, a challenging benchmark where many established tools performed near the level of random chance. The performance advantage of Hi-TPH-PLMs is also attributed to the on-the-fly selection of nonbinding data for learning a resilient decision boundary and preventing overfitting to dataset artifacts. These results underscore a critical shift from pursuing pure algorithmic complexity to leveraging the synergy between high-quality data and foundational PLMs.

Leveraging the hierarchical design of Hi-TPH, our analysis reveals that contributions of prediction components are synergistic rather than simply additive. In particular, while the TCR CDR3α sequence plays an indispensable role, the contribution of the HLA sequence is highly context dependent, emerging only when full TCR sequences are provided. An attention map analysis further validates that the component contributions in Hi-TPH-PLMs are consistent with the established biophysical model of TCR–pHLA recognition. These findings underscore the necessity of a holistic representation of the TCR–pHLA interface to capture the synergistic interplay among its components, which further reflects the potential of large PLMs in this binding prediction task.

Hi-TPH and Hi-TPH-PLMs establish a foundational platform for immunoinformatics research, which provides a validated toolkit adaptable to the varied data completeness of real-world research, from basic peptide/CDR3β sequences to full TCR sequences. The platform can be integrated into versatile frameworks like BioSeq-BLM[31,32], which would allow its models and data to be applied to a broader range of tasks. This approach of combining high-quality data with advanced models redefines performance standards and offers a clear path toward building reliable, clinically useful prediction tools for personalized immunotherapy.

In the future, Hi-TPH could be refined by adding quantitative binding affinity data to build models for therapeutic applications that require a nuanced understanding of interaction strengths. It can also be improved to include HLA class II for immediate applicability to $CD4^+$

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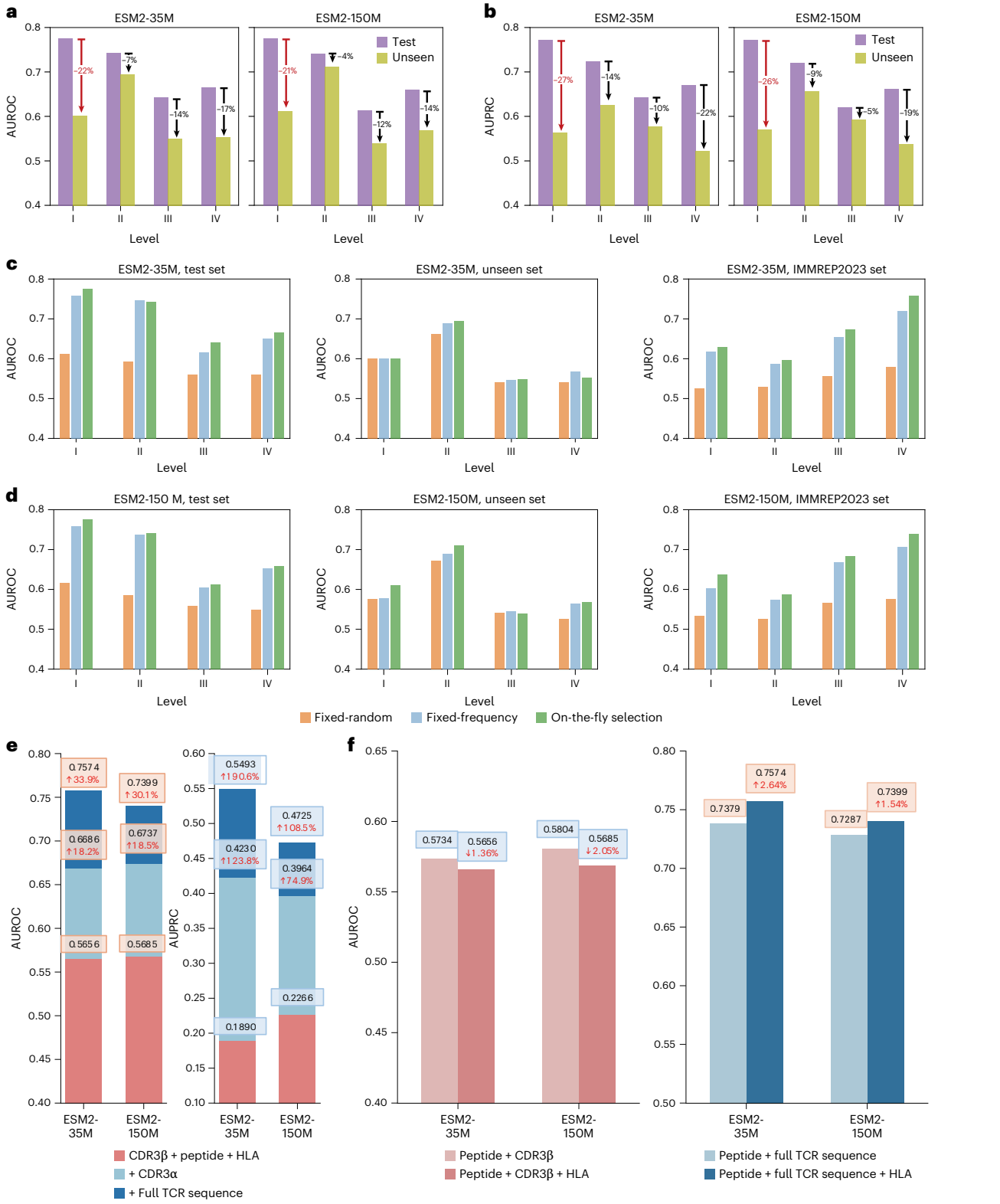


Fig. 4 | Benchmarking results of unseen peptide generalization, on-the-fly selection of nonbinding data, and component contributions using Hi-TPH-PLMs. a,**b**, Peptide generalization of Hi-TPH-PLMs. Shown are AUROC (**a**) and AUPRC (**b**) scores for ESM2-35M and ESM2-150M models when predicting binding on test set versus unseen set across Hi-TPH Levels I-IV. **c**,**d**, Efficacy of the on-the-fly selection for nonbinding data generation. AUROC scores for the ESM2-35M (**c**) and ESM2-150M (**d**) are evaluated across the test, unseen and IMMREP2023 sets. The on-the-fly selection method is benchmarked against fixed-random and fixed-frequency methods. **e**, Incremental contribution of TCR components to model performance on the IMMREP2023 set. Stacked bars show AUROC improvements for ESM2-35M and ESM2-150M as CDR3α and full TCR sequences are added to models using peptide, HLA and CDR3β. Percentages indicate the relative gain contributed by each additional component. **f**, Context-dependent contribution of HLA to model performance on the IMMREP2023 set by comparing models with and without HLA sequences for two different TCR input sequences (peptide + CDR3β, left versus peptide + full TCR sequence, right).

T cell responses. Hi-TPH exhibits biases from original data sources such as the uneven distribution of sequence length and HLA alleles, leading Hi-TPH-trained models to favor more common lengths (9-mer peptides and 15-residue CDR3s; Supplementary Fig. 9) or certain HLA alleles (Supplementary Table 7 and Supplementary Fig. 10). To achieve broader applicability, it is worth exploring strategies to mitigate the impact of these biases.

In addition, the performance of Hi-TPH-PLMs scales with model size only when the input data are sufficiently complex (Supplementary Figs. 2 and 3), which necessitates further investigation to transfer the benefit of continuously advancing PLMs to the TCR–pHLA binding prediction task. Hi-TPH-PLMs can further serve as efficient first-stage screening models, narrowing large candidate spaces before computationally intensive structure-based ranking[33]. We also evaluated existing tools and Hi-TPH-PLMs on the recently released IMMREP2025[34] benchmark (Supplementary Table 17). IMMREP2025 designs a more challenging setting: for each positive TCR–pHLA pair, negative samples are generated by pairing the same TCR with alternative peptides within the same HLA, and the peptides are deliberately distant from published epitopes. These choices reduce peptide-side motif similarities available to sequence-based models, leading to lower overall performance and smaller method differences compared with IMMREP2023. Consistent with the IMMREP2025 report[34], where top-performing methods incorporated structural modeling, fine-grained peptide specificity prediction may benefit from integration with folding models and structure-based rescoring. Leveraging Hi-TPH to construct tailored training datasets that capture subtle peptide-dependent binding patterns represents another promising direction for improving Hi-TPH-PLMs[35].

Methods

The Hi-TPH dataset construction

Binding data collection. The Hi-TPH dataset was constructed by collecting positive (binding) TCR–pHLA interaction data from two distinct sources, complemented by a sequence reconstruction pipeline to ensure comprehensive coverage of available binding evidence. For source 1 (biological database collection), primary data were aggregated from four immunological repositories, including the Immune Epitope Database[24] (IEDB; April 2024 access), VDJdb[25] (2023-06-01 version), McPAS-TCR[26] (2022-09-10 version) and ImmuneCODE-MIRA[27] (release-002.1 version), where IEDB data were restricted to linear peptides with positive T cell assays in human hosts, and ImmuneCODE-MIRA entries were deduplicated to retain only unique peptide–TCR combinations. For source 2 (published literature integration), additional TCR–pHLA binding pairs were extracted from peer-reviewed publications, incorporating datasets associated with established computational tools, including PanPep[18], pMTnet[13], TEIM[36], epiTCR[17] and STAPLER[15]. For sequence reconstruction, to comprehensively expand dataset coverage beyond the collected amino acid sequences, we processed V(D)J gene annotations through a reconstruction pipeline to generate full TCR variable region sequences. As illustrated in Fig. 2b, this pipeline uses an assembly approach based on standardized gene definitions. The process begins by cross-referencing V and J gene annotations with the IMGT/Gene-DB repository[37] to retrieve the corresponding germline amino acid sequences. The experimentally determined CDR3 sequence is then extracted to serve as the hypervariable junction. By precisely truncating the germline V and J sequences at conserved recombination signal motifs and subsequently concatenating them with the CDR3 region, we reconstruct the full variable regions of both TCR α and β chains. As compared with existing tools like Stitchr[38] for full reconstruction, our pipeline omits the constant region that is meaningless for prediction models to discriminate binding pairs from nonbinding ones. In addition, our pipeline maintains high fidelity by directly querying the latest reference databases, which prevents potential sequence mismatches (Supplementary Fig. 6).

Data curation and hierarchical levels. Following binding data collection, all records underwent a multistep data curation pipeline to ensure quality, consistency and suitability for machine learning applications. First, we performed data standardization and conflict resolution. Records from different sources were mapped to a unified schema. Next, we applied a series of component-specific filters to focus the dataset on the most common and well-characterized interaction types while removing potential outliers that could interfere with model training.

- HLA type: Only records associated with HLA class I molecules were retained.
- Peptide length: We restricted the dataset to peptides of 8–15 amino acids in length, aligning with the canonical length distribution of naturally presented peptides.
- CDR3 length: Similarly, CDR3 sequences were filtered to lengths of 9–19 amino acids, consistent with the dominant length distribution observed in human TCR repertoires.
- Full TCR chain length: For the reconstructed sequences, we imposed length constraints of 105–121 (TCR α) and 109–121 (TCR β) amino acids to eliminate potential reconstruction artifacts or outliers.

This rigorous curation process resulted in a clean, high-quality master dataset of binding interactions. From this unified resource, we defined four hierarchical levels to enable the systematic dissection of component contributions (Fig. 1a). Each level represents a progressively richer set of molecular information:

- Level I: represents the minimal component combination, consisting only of the peptide and TCR CDR3β.
- Level II: extends level I by including HLA sequences.
- Level III: incorporates the peptide, HLA and both the TCR CDR3α and CDR3β sequences.
- Level IV: the most comprehensive level, covering the peptide, HLA and the full TCR α and β variable sequences.

This hierarchical structure is a core feature of Hi-TPH, shaping Hi-TPH into four subsets named Hi-TPH levels I–IV with increasing prediction components and decreasing sizes.

On-the-fly selection method for nonbinding data generation. Existing TCR–pHLA binding prediction tools often rely on static negative datasets repeatedly used across all training epochs[5], which may introduce systematic biases and fail to adequately represent the vast nonbinding interaction space. Fixed-random and fixed-frequency are widely used methods for generating an immutable set of nonbinding pairs before training. The fixed-random approach constructs this set via combinatorial pairing of unique peptides and TCRs, excluding known positives[7,18,36,39,40]. The fixed-frequency method generates nonbinding data by sampling peptides proportional to their frequency in the positive dataset and pairing them with randomly selected TCRs, while strictly excluding known positive pairs. This ensures that the peptide distribution in the negative set mirrors that of the positive class[40,41].

The on-the-fly selection method generates negative samples dynamically along the training process according to positive TCR–pHLA pairs at each training step. For each peptide within a positive TCR–pHLA pair, a negative instance is generated by pairing the peptide with a randomly selected TCR from the training set. Note that any generated pair that matches a known binding pair in the training set is discarded (Supplementary Algorithm 1). This dynamic approach ensures diversity across epochs and preserves the peptide frequency distribution observed in the positive dataset[29]. Diversity across epochs will expand the coverage of peptides and TCRs, which is critical for the learning of prediction models (Supplementary Table 4).

Benchmarking experimental setup

Task definition. The TCR–pHLA binding prediction task is formulated as a binary classification problem to predict whether a given TCR binds to a specific pHLA complex. Hi-TPH not only provides positive binding pairs but also incorporates an on-the-fly selection method of nonbinding data generation to create balanced training and evaluation datasets.

Dataset splitting. The dataset splitting strategy was designed to assess model performance under different generalization scenarios. Peptides in the dataset were first ranked by their frequency of occurrence, creating a distribution from high-frequency head peptides to low-frequency long-tail peptides. In this study, peptides with fewer than 50 binding samples were regarded as long-tail peptides. To evaluate generalization to novel peptides, we randomly selected 100 peptides from the long-tail peptide pool and constructed an unseen set by holding out all samples associated with these peptides, ensuring no peptide overlap with any of the remaining partitions. The remaining data were then randomly split into training, validation, and the internal test set at an 8:1:1 ratio for standard held-out evaluation. This splitting strategy yields three evaluation settings: the internal test set for general performance assessment, the unseen set for evaluating generalization to unseen peptides, and the IMMREP2023 set[23] for external validation.

Overlap removal for external evaluation on IMMREP2023. To reduce potential overlap between the external IMMREP2023 benchmark and the data used for model development, we explicitly screened for overlapping TCR–pHLA pairs between IMMREP2023 and the Hi-TPH training and validation sets at each level (levels I–IV). Because different Hi-TPH levels use different pair representations, overlap screening was performed separately for each level, and the resulting overlap statistics are summarized in Supplementary Table 13. Any overlapping pairs identified in the corresponding Hi-TPH training and validation sets were removed before model training and evaluation. All results reported on IMMREP2023 were obtained after this overlap-removal procedure. The impact of this filtering was further quantified by comparing IMMREP2023 performance before and after retraining on the filtered datasets (Supplementary Table 14). In addition, as shown in Supplementary Fig. 8, IMMREP2023 contains a high proportion of peptides and TCRs that are unseen in the Hi-TPH training sets, making it a challenging benchmark for assessing real-world generalization.

Evaluation metrics. Model performance was evaluated using two primary metrics: the AUROC and the area under the precision-recall curve (AUPRC). AUROC measures a model's ability to distinguish between positive and negative classes by plotting the true positive rate against the false positive rate across all classification thresholds. AUPRC evaluates performance by plotting precision against recall at various thresholds, providing insight into model performance on imbalanced datasets. For the internal test and unseen sets, we report the mean metrics across five independent runs, each utilizing distinct randomly sampled negatives to maintain a strict 1:1 ratio. The IMMREP2023 set was evaluated directly on its inherently imbalanced class distribution without additional sampling.

Input representation and model implementation

Input representation. To prepare the sequence data for Hi-TPH-PLMs, we developed a standardized input pipeline. The input for each model training on a given Hi-TPH level is constructed by concatenating the corresponding prediction components into a single sequence. To accommodate the fixed-size input requirements of the transformer architectures, all sequences were first normalized to a uniform length through padding with a designated special character (X). As compared with concatenating representations after passing each component through the PLM individually like TCR-ESM[42], sequence-level concatenation facilitates more effective exploitation of PLMs' capabilities (Supplementary Table 8 and Supplementary Fig. 7).

Model implementation. Hi-TPH-PLMs contain a pretrained PLM serving as a feature extractor (the 'backbone') and a multilayer classifier (the 'head'), which are jointly optimized in the training phase (Fig. 3).

- PLM backbone: We selected a diverse suite of PLMs as the backbone to systematically evaluate the impact of model scale, architecture and pretraining data on prediction performance. Our selection spanned the evolution of protein transformers, from early models like TAPE-BERT[43] and ProtBERT[44] to the widely adopted ESM2 family[45], including variants ranging from 8M to 650M parameters to assess the impact of model scaling (Supplementary Figs. 2 and 3). To benchmark against the latest advancements, we incorporated AMPLIFY-120M-base, AMPLIFY-120M and AMPLIFY-350M from the AMPLIFY family[46]. These models integrate modern architectural improvements, such as SwiGLU[47], RMSNorm[48] and FlashAttention[49], and were pretrained on a vast corpus of protein sequences including UniRef100[50], OAS[51] and SCOP[52].
- Classification head: To adapt these powerful feature extractors for our specific binary classification task, the classification head was appended to the final hidden state of each PLM. This projection head consisted of two fully-connected hidden layers with a rectified linear unit (ReLU) activation function, which processed the learned protein representations to produce the final binding prediction.

Model ensemble and metric averaging. For each Hi-TPH level, Hi-TPH-PLMs-Ens. was constructed by averaging the prediction scores of all individual Hi-TPH-PLMs reported in Table 2 (ESM2-8M/35M/150M/650M, TAPE-BERT, AMPLIFY-120M-base/120M/350M and ProtBERT) with equal weights. Formally, for an input pair x at level ℓ, the ensemble prediction $f_{\mathrm{Ens}}^{(\ell)}(x)$ is defined as

$$f_{\mathrm{Ens}}^{(\ell)}(x) = \frac{1}{M_\ell} \sum_{m=1}^{M_\ell} f_m^{(\ell)}(x),$$

where $f_m^{(\ell)}(x)$ denotes the predicted binding score from the mth Hi-TPH-PLM at level ℓ, and M_ℓ is the number of individual models included at that level. No model selection, weighting or hyperparameter tuning was performed on IMMREP2023 for the ensemble. We additionally compared this full ensemble with different ensemble strategies on IMMREP2023 and the held-out Hi-TPH test set (Supplementary Tables 15 and 16).

Hi-TPH-PLMs-Avg. is a metric-level reference summary rather than an additional prediction model or model ensemble. For each evaluation metric s (for example, AUROC or AUPRC), it is calculated as

$$s_{\mathrm{Avg}}^{(\ell)} = \frac{1}{M_\ell} \sum_{m=1}^{M_\ell} s_m^{(\ell)},$$

where $s_m^{(\ell)}$ denotes the performance metric of the mth individual Hi-TPH-PLM at level ℓ.

Model training procedures

All Hi-TPH-PLMs were fine-tuned using the Adam optimizer[53] with a cross-entropy loss function. The training was implemented in PyTorch[54] and distributed across four NVIDIA RTX 3090 graphics processing units to ensure feasible training times. To prevent overfitting and select the optimal model checkpoint for evaluation, we used an early stopping protocol. After each training epoch, model performance was evaluated on the validation set using AUROC as the primary metric. The training process was terminated if the validation AUROC failed to improve for four consecutive epochs (a patience

of 4). The model checkpoint corresponding to the epoch with the highest validation AUROC was saved and used for all subsequent evaluations on the test sets. The batch size was set to 32 per graphics processing unit for most models. To accommodate the memory requirements of larger backbones, it was reduced to 8 for ESM2-150M, AMPLIFY-350M and ProtBERT, and further to 2 for ESM2-650M at level IV. The reduction of batch size has neglectable influence on the performance of these models (Supplementary Table 10). The learning rate was selected from {1e−4, 8e−5, 5e−5, 3e−5, 1e−5} based on validation AUROC.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The Hi-TPH dataset is publicly available via GitHub at https://github.com/Jiadong001/Hi-TpH and via Zenodo at https://doi.org/10.5281/zenodo.22647655 (ref. 55), licensed under the Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0) license. Source data are provided with this paper.

Code availability

The code for using Hi-TPH, as well as the benchmark, is available via GitHub at https://github.com/Jiadong001/Hi-TpH and via Zenodo at https://doi.org/10.5281/zenodo.22647655 (ref. 55).

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Author contributions

X.Z., J.L. and F.F. conceived the study. J.L. collected and processed the data. J.L., Y.L. and Y.Z. performed the experiments. X.Z. and J.L. analyzed the results, prepared the figures and wrote the original manuscript. F.F. reviewed and revised the manuscript. All authors read and approved the manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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Software and code

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Data collection	We used python 3.8 with pandas 2.0.3, numpy 1.24.3, biopython 1.83, and matplotlib 3.7.2 for data collection. The code can be obtained at https://github.com/Jiadong001/Hi-TpH.
Data analysis	We used python 3.8 with pandas 2.0.3, numpy 1.24.3, biopython 1.83, and matplotlib 3.7.2 for data analysis. The code can be obtained at https://github.com/Jiadong001/Hi-TpH.

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All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

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- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

The raw data is collected from https://www.iedb.org/, https://github.com/antigenomics/vdjdb-db/releases/tag/2023-06-01, https://friedmanlab.weizmann.ac.il/

McPAS-TCR, and https://clients.adaptivebiotech.com/pub/covid-2020. The IMMREP2023 test set is obtained at https://github.com/justin-barton/IMMREP23/. The prepared benchmarking set is available at https://github.com/Jiadong001/Hi-TpH.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	n/a
Reporting on race, ethnicity, or other socially relevant groupings	n/a
Population characteristics	n/a
Recruitment	n/a
Ethics oversight	n/a

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	The datasets were selected based on the availability of relevant TCR–pHLA binding data and the need to evaluate different levels of molecular information and generalization. Hi-TPH integrates data from four public databases (IEDB, VDJdb, McPAS-TCR and ImmuneCODE) and five published collections. Following curation, the data were organized into four hierarchical subsets containing 214,641, 78,679, 28,262 and 26,704 binding pairs, respectively. Evaluation used an internal held-out Test set, an Unseen set comprising held-out peptides, and the external IMMREP2023 set. The scale and molecular diversity of Hi-TPH support model training and controlled component analyses, while the complementary evaluation settings assess performance on held-out interactions, unseen peptides and external data.
Data exclusions	In benchmark, we limited the peptide length to 8-15 for level I and 8-10 for levels II-IV, the CDR3 length to 9-19 for levels I-IV, and the full TCRα and β sequence lengths to 105-121 and 109-121, respectively, for level IV. Samples that did not meet these requirements were eliminated.
Replication	Model Training and model evaluation were performed five times with five different random seeds, and the results were consistent. All attempts at replication were successful.
Randomization	Allocation to experimental groups was not applicable because this study involved computational analyses of existing datasets, with no experimental subjects or treatment groups. For model development and evaluation, 100 peptides were randomly selected from the long-tail peptide pool, and all associated samples were held out as the Unseen set. The remaining data were randomly divided into training, validation and test sets at an 8:1:1 ratio.
Blinding	Blinding to group allocation was not applicable because this study involved computational analyses of existing datasets, with no experimental subjects or treatment-group allocation. No new wet-lab data were collected.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	Not applicable to the current study.
Novel plant genotypes	Not applicable to the current study.
Authentication	Not applicable to the current study.